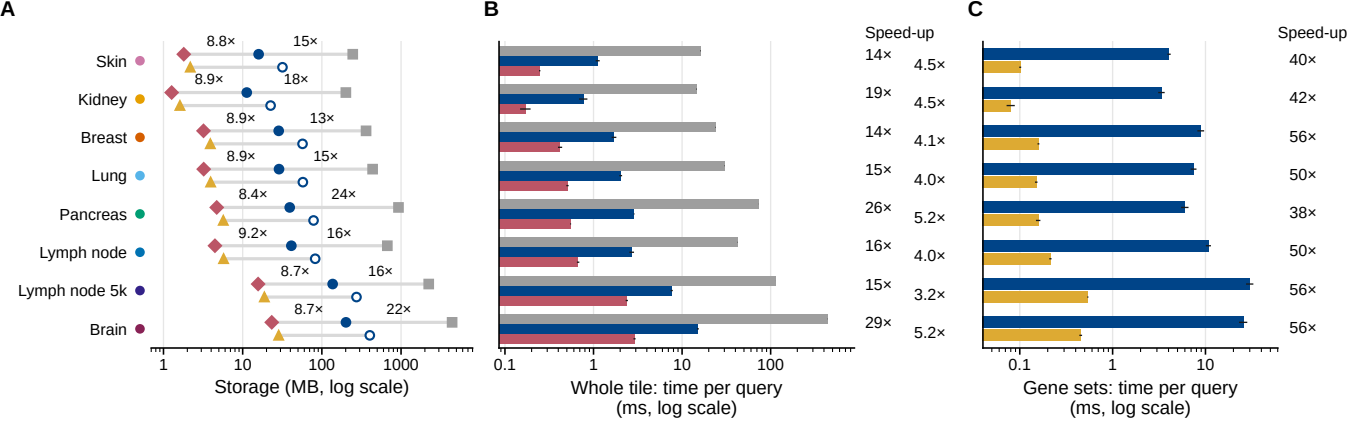
A
B
C
Skin
Kidney
Breast
Lung
Pancreas
Lymph node
Lymph node 5k
Brain
8.8×
15×
8.9×
18×
8.9×
13×
8.9×
15×
8.4×
24×
9.2×
16×
8.7×
16×
8.7×
22×
1
10
100
1000
Storage (MB, log scale)
Speed-up
14×
19×
14×
15×
26×
16×
15×
29×
0.1
1
10
100
Whole tile: time per query
(ms, log scale)
Speed-up
4.5×
4.5×
4.1×
4.0×
5.2×
4.0×
3.2×
5.2×
40×
42×
56×
50×
38×
50×
56×
56×
0.1
1
10
Gene sets: time per query
(ms, log scale)
Whole covariances
Spindle-naive
Spindle
Spindle-naive (gene sets)
Spindle (gene sets)
Whole-matrix scan
Spindle-naive
Spindle
Spindle (gene sets)